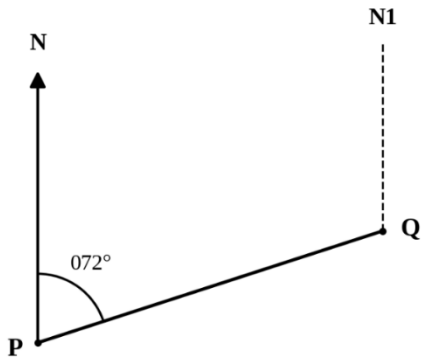
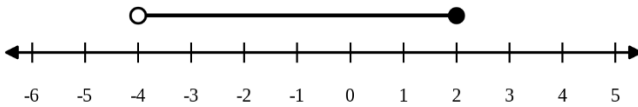
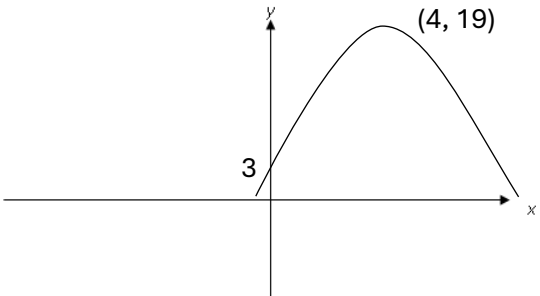


2026 Mathematics Paper 1 Solutions

Q	Solutions
1a	1.56×10^{-4}
1b	7.65×10^{150}
2	 $\angle N_1PQ = 108^\circ$ (int. angles) Bearing $= 360^\circ - 108^\circ$ (angles at a pt) $= 252^\circ$
3	$8^{y+1} = 32^3$ $2^{3y+3} = 2^{15}$ $3y + 3 = 15$ $3y = 12$ $y = 4$
4	$\frac{x}{2} = 36.8699$ or 143.1301 $x = 73.7$ or 286.3 (1 d.p.)
5	$(2x-3)(x+5) - (x-1)^2$ $= 2x^2 + 7x - 15 - (x^2 - 2x + 1)$ $= x^2 + 9x - 16$
6	Let T be the total number of cards. Common cards $= 0.34T$ Remaining cards $= 0.66T$ Legendary cards $= 0.4(0.66T)$ $0.4(0.66T) = 132$ $0.264T = 132$ $T = 500$ Alternative Method

	$\frac{33}{125} \text{ units} - 132$ $1 \text{ unit} - 132 \div \frac{33}{125}$ $1 \text{ unit} = 500$
7a	$180(n-2) = 2340$ $180n - 360 = 2340$ $180n = 2700$ $n = 15$
7b	$\frac{360}{15} = 24^\circ$
8a	$p = \frac{2x+8}{x+1}$ $p(x+1) = 2x+8$ $px + p = 2x+8$ $px - 2x = 8 - p$ $x(p-2) = 8 - p$ $x = \frac{8-p}{p-2}$
8b	$x = \frac{8-3}{3-2}$ $x = 5$
9	

10	$I = \frac{k}{d^2}$ $I_{new} = \frac{k}{(1.6d)^2} = \frac{k}{2.56d^2} = 0.390625I$ $\frac{I - 0.390625I}{I} \times 100\% = 60.9\% \text{ (3 s.f.)}$
11a	1.05 seconds
11b	$\frac{3.5 + 3.c}{2} = 3.65$ $3.5 + 3.c = 7.3$ $3.c = 3.8$ $c = 8$
12a	$\sqrt{\frac{4}{1}} = 2:1$ $\therefore 1:2$
12b	Mass Ratio = 1:8 Large mass = $300 \times 8 = 2400g$
13a	$(P \cup Q) \cap (P \cap Q)'$
13bi	A is the set of positive integers that are factors of 10 .
13bii	$\{-1, 0, 4, 6, 8, 9, 12, 14, 15\}$
13biii	6
14a	$\frac{x+2}{4} \leq 5 - 2x < 13$ $x+2 \leq 20 - 8x \text{ and } -2x < 8$ $\therefore -4 < x \leq 2$
14b	
15a	$\angle ACB$ $= 180^\circ - 50^\circ - 65^\circ \text{ (}\angle \text{ sum of } \Delta \text{)}$ $= 65^\circ$ $\angle ADB = 65^\circ \text{ (}\angle \text{s in the same segment)}$

15b	$\angle ABD$ $= 180^\circ - 26^\circ - 65^\circ$ ($\angle$ sum of Δ) $= 89^\circ$ Since $89^\circ \neq 90^\circ$, by converse of angle in a semicircle property, AD does not pass through the centre of the circle.
16a	1 : 40000
16b	30 cm
16c	1 cm : 0.4 km $1 \text{ cm}^2 : 0.16 \text{ km}^2$ Actual area $= 8 \times 0.16$ $= 1.28 \text{ km}^2$
17a	$y = -(x^2 - 8x) + 3$ $y = -[(x - 4)^2 - 16] + 3$ $y = -(x - 4)^2 + 19$
17b	
18a	$\frac{14}{23}$
18b	$\frac{3n - 1}{5n - 2}$
18c	$\frac{3n - 1}{5n - 2} = \frac{503}{800}$ $800(3n - 1) = 503(5n - 2)$ $206 = 115n$ $n = 1.79...$ As n is not a positive integer, $\frac{503}{800}$ is not a term of the sequence.
19a	$(3x - 4)(2x + 3)$

19b	$6x^2 - 9x + 4xy - 6y$ $= 3x(2x - 3) + 2y(2x - 3)$ $= (3x + 2y)(2x - 3)$
20a	The first year's interest is more than 8% of \$12 000 because the interest is compounded every quarter , so the principal amount increases every quarter .
20b	$A = (12000) \left(1 + \frac{8 \div 4}{100} \right)^{4 \times 4}$ $I = (12000) \left(1 + \frac{8 \div 4}{100} \right)^{4 \times 4} - 12000$ $I = 4473.43 \text{ (2 d.p.)}$
21	$6p + q = 106$ $q - 2p = 58$ By substitution or elimination, $p = 6, q = 70$
22a	$y = \frac{5}{2}x + c$ $2 = \frac{5}{2}(9) + c$ $c = 2 - \frac{45}{2}$ $c = -\frac{41}{2}$ $y = \frac{5}{2}(11) - \frac{41}{2}$ $y = 7$ $\therefore R(11, 7)$
22b	$(3, 7)$
22c	40 units ²
23a	$216 = 2^3 \times 3^3$
23b	24,108
23c	$1728 \div \frac{p^2}{q}$ $= 2^6 \times 3^3 \times \frac{q}{p^2}$ $p = 2, q = 3$
24a	1.5 m/s ²
24b	Greater. The speed change is the same (0 to 12 m/s) but occurs over a shorter time (6 s vs 8 s), giving a larger rate of change .

24c	Motorbike's distance travelled during T seconds $= 48 + 12(t - 8)$ Cyclist's distance travelled during T seconds $= 12 + 6t$ $12 + -48 = 12 + 6t$ $t = 10$
25ai	$\vec{AB} = b - a$
25aii	$\vec{OC}$ $= a + \frac{3}{4}(b - a)$ $= \frac{1}{4}a + \frac{3}{4}b$
25aiii	$\vec{OD}$ $= \frac{3}{7}b$
25bi	$\vec{AD}$ $= \vec{OD} - \vec{OA}$ $= \frac{3}{7}b - a$ $= -a + \frac{3}{7}b$ $\vec{AQ}$ $= \vec{OQ} - \vec{OA}$ $= \frac{1}{8}a + \frac{3}{8}b - a$ $= -\frac{7}{8}a + \frac{3}{8}b$ $= \frac{7}{8}\left(-a + \frac{3}{7}b\right)$ $= \frac{7}{8}\vec{AD}$ As $\vec{AQ} = \frac{7}{8}\vec{AD}$ and A is a common point, A, Q and D are collinear.
25bii	$AQ : QD = 7 : 1$

